13 Engineering Mathematics: Discrete Mathematics (36)

13.0.1 UGC NET CSE | June 2012 | Part 3 | Question: 17

Let $Q(x, y)$ denote "x+y=0" and let there be two quantifications given as

I. $\exists y \forall x Q(x, y)$
II. $\forall x \exists y Q(x, y)$

where x and y are real numbers. Then which of the following is valid?

A. I is true and II is false
B. I is false and II is true
C. I is false and II is also false
D. both I and II are true

ugcnetcse-june2012-paper3 discrete-mathematics mathematical-logic

Answer key

13.0.2 UGC NET CSE | June 2012 | Part 3 | Question: 50

How many relations are there on a set with n elements that are symmetric and a set with n elements that are reflexive and symmetric?

A. $2^{n(n+1)/2}$ and $2^n.3^{n(n-1)/2}$
B. $3^{n(n-1)/2}$ and $2^{n(n-1)}$
C. $2^{n(n+1)/2}$ and $3^{n(n-1)/2}$
D. $2^{n(n+1)/2}$ and $2^{n(n-1)/2}$

ugcnetcse-june2012-paper3 discrete-mathematics set-theory&algebra

Answer key

13.0.3 UGC NET CSE | December 2014 | Part 2 | Question: 01

Consider a set $A = \{1, 2, 3, \dots\dots\dots, 1000\}$.

How many members of A shall be divisible by 3 or by 5 or by both 3 and 5?

A. 533
B. 599
C. 467
D. 66

ugcnetcse-dec2014-paper2 discrete-mathematics set-theory&algebra

Answer key

13.0.4 UGC NET CSE | December 2013 | Part 3 | Question: 24

The objective of ____ procedure is to discover at least one ____ that causes two literals to match.

A. unification, validation
B. unification, substituion
C. substitution, unification
D. minimax, maximum

ugcnetcse-dec2013-paper3 discrete-mathematics mathematical-logic

Answer key

13.0.5 UGC NET CSE | Junet 2015 | Part 3 | Question: 24

Which one of the following is true?

A. The resolvent of two Horn clauses is not a Horn clause
B. The resolvent of two Horn clauses is a Horn clause
C. If we resolve a negated goal G against a fact or rule A to get clause C then C has positive literal or non-null goal
D. If we resolve a negated goal G against a fact or rule A to get clause C then C has positive literal or null goal

ugcnetcse-june2015-paper3 discrete-mathematics mathematical-logic

Answer key

13.0.6 UGC NET CSE | July 2018 | Part 2 | Question: 86

If $A_i = \{-i, \ldots, -2, -1, 0, 1, 2, \ldots, i\}$ then $\cup_{i=1}^{\infty} A_i$ is

A. Z B. Q C. R D. C

ugcnetcse-july2018-paper2 discrete-mathematics

Answer key

13.0.7 UGC NET CSE | October 2020 | Part 2 | Question: 86

Let G be a simple undirected graph, T_D be a DFS tree on G, and T_B be the BFS tree on G. Consider the following statements.

Statement I: No edge of G is a cross with respect to T_D

Statement II: For every edge (u, v) of G, if u is at depth i and v is at depth j in T_B then $|i - j| = 1$

In the light of the above statements, choose the correct answer from the options given below

A. Both Statement I and Statement II are true
B. Both Statement I and Statement II are false
C. Statement I is correct but Statement II is false
D. Statement I is incorrect but Statement II is true

ugcnetcse-oct2020-paper2 discrete-mathematics graph-theory

13.0.8 UGC NET CSE | October 2020 | Part 2 | Question: 38

What kind of clauses are available in conjunctive normal form?

A. Disjunction of literals
B. Disjunction of variables
C. Conjunction of literals
D. Conjunction of variables

ugcnetcse-oct2020-paper2 discrete-mathematics mathematical-logic

Answer key

13.0.9 UGC NET CSE | October 2020 | Part 2 | Question: 37

If $f(x) = x$ is my friend, and $p(x) = x$ is perfect, then correct logical translation of the statement "some of my friends are not perfect" is ______

A. $\forall_x (f(x) \wedge \neg p(x))$
B. $\exists_x (f(x) \wedge \neg p(x))$
C. $\neg(f(x) \wedge \neg p(x))$
D. $\exists_x (\neg f(x) \wedge \neg p(x))$

ugcnetcse-oct2020-paper2 discrete-mathematics mathematical-logic

Answer key

13.0.10 UGC NET CSE | October 2020 | Part 2 | Question: 26

Let G be a directed graph whose vertex set is the set of numbers from 1 to 100. There is an edge from a vertex i to a vertex j if and only if either $j = i + 1$ or $j = 3i$. The minimum number of edges in a path in G from vertex 1 to vertex 100 is ______

A. 23 B. 99 C. 4 D. 7

ugcnetcse-oct2020-paper2 discrete-mathematics graph-theory

Answer key

13.0.11 UGC NET CSE | October 2020 | Part 2 | Question: 3

Which of the following pairs of propositions are not logically equivalent?

A. $((p \rightarrow r) \wedge (q \rightarrow r))$ and $((p \vee q) \rightarrow r)$

B. $p \leftrightarrow q$ and $(\neg p \leftrightarrow \neg q)$
C. $((p \wedge q) \vee (\neg p \wedge \neg q))$ and $p \leftrightarrow q$
D. $((p \wedge q) \rightarrow r)$ and $((p \rightarrow r) \wedge (q \rightarrow r))$

ugcnetcse-oct2020-paper2 discrete-mathematics mathematical-logic

Answer key

13.1 Boolean Function (1)

13.1.1 Boolean Function: UGC NET CSE | August 2016 | Part 2 | Question: 1

The Boolean function $[\sim (\sim p \wedge q) \wedge \sim (\sim p \wedge \sim q)] \vee (p \wedge r)$ is equal to the Boolean function :

A. q B. $p \wedge r$ C. $p \vee q$ D. p

ugcnetcse-aug2016-paper2 discrete-mathematics boolean-function

Answer key

13.2 Combinatory (2)

13.2.1 Combinatory: UGC NET CSE | December 2012 | Part 3 | Question: 21

How many solutions do the following equations have?

$x_1 + x_2 + x_3 = 11$

where $x_1 \geq 1, x_2 \geq 2, x_3 \geq 3$

A. $C(7,11)$ B. $C(11,3)$ C. $C(14,11)$ D. $C(7,5)$

ugcnetcse-dec2012-paper3 discrete-mathematics combinatory

Answer key

13.2.2 Combinatory: UGC NET CSE | December 2015 | Part 2 | Question: 48

How many solutions are there for the equation $x + y + z + u = 29$ subject to the constraints that $x \geq 1,\ \ y \geq 2\ \ z \geq 3\ \ and\ \ u \geq 0$?

A. 4960 B. 2600 C. 23751 D. 8855

ugcnetcse-dec2015-paper2 discrete-mathematics combinatory

Answer key

13.3 Equivalence Class (1)

13.3.1 Equivalence Class: UGC NET CSE | July 2018 | Part 2 | Question: 89

Which of the following is an equivalence relation on the set of all functions from Z to Z?

A. $\{f,\ g) \mid f(x) - g(x) = 1 \ \forall\, x \in\ Z\}$
B. $\{f,\ g) \mid f(0) = g(0)$ or $f(1) = g(1)\}$
C. $\{f,\ g) \mid f(0) = g(1)$ and $f(1) = g(0)\}$
D. $\{f,\ g) \mid f(x) - g(x) = k$ for some $k \in Z\}$

ugcnetcse-july2018-paper2 discrete-mathematics equivalence-class

Answer key

13.4 First Order Logic (2)

13.4.1 First Order Logic: UGC NET CSE | December 2012 | Part 3 | Question: 58

Skolmization is the process of

A. bringing all the quantifiers in the beginning of a formula in FDL
B. removing all the universal quantifiers
C. removing all the extential quantifiers
D. all of the above

ugcnetcse-dec2012-paper3 engineering-mathematics discrete-mathematics first-order-logic

Answer key

13.4.2 First Order Logic: UGC NET CSE | October 2020 | Part 2 | Question: 40

Consider the following argument with premise $\forall_x(P(x) \vee Q(x))$ and conclusion $(\forall_x P(x)) \wedge (\forall_x Q(x))$

(A) $\forall_x(P(x) \vee Q(x))$	Premise
(B) $P(c) \vee Q(c)$	Universal instantiation from (A)
(C) $P(c)$	Simplification from (B)
(D) $\forall_x P(x)$	Universal Generalization of (C)
(E) $Q(c)$	Simplification from (B)
(F) $\forall_x Q(x)$	Universal Generalization of (E)
(G) $(\forall_x P(x)) \wedge (\forall_x Q(x))$	Conjuction of (D) and (F)

A. This is a valid argument
B. Steps (C) and (E) are not correct inferences
C. Steps (D) and (F) are not correct inferences
D. Step (G) is not a correct inference

ugcnetcse-oct2020-paper2 discrete-mathematics first-order-logic

Answer key

13.5 Functions (2)

13.5.1 Functions: UGC NET CSE | December 2011 | Part 2 | Question: 4

Domain and Range of the function $Y = -\sqrt{-2x+3}$ is

A. $x \geq \frac{3}{2}, y \geq 0$
B. $x > \frac{3}{2}, y \leq 0$
C. $x \geq \frac{3}{2}, y \leq 0$
D. $x \leq \frac{3}{2}, y \leq 0$

ugcnetcse-dec2011-paper2 discrete-mathematics functions

Answer key

13.5.2 Functions: UGC NET CSE | December 2012 | Part 3 | Question: 19

Identify the following activation function:

$\Phi(V) = Z + \frac{1}{1+exp(-x*V+Y)}$, Z, X, Y are parameters.

A. Step function
B. Ramp function
C. Sigmoid function
D. Gaussian function

ugcnetcse-dec2012-paper3 discrete-mathematics functions

Answer key

13.6 Group Theory (1)

13.6.1 Group Theory: UGC NET CSE | June 2012 | Part 3 | Question: 67

Let a*H and b*H be two cosets of H.

I. Either a*H and b*H are disjoint
II. a*H and b*H are identical

Then,

A. Only I is true
B. Only II is true
C. I or II is true
D. I and II is false

ugcnetcse-june2012-paper3 discrete-mathematics set-theory&algebra group-theory

Answer key

13.7 Linear Programming (1)

13.7.1 Linear Programming: UGC NET CSE | July 2016 | Part 3 | Question: 70

Consider the statement

"Either $-2 \leq x \leq -1$ or $1 \leq x \leq 2$"

The negation of this statement is

A. x<-2 or 2<x or -1<x<1
B. x<-2 or 2<x
C. -1<x<1
D. x $\leq$ -2 or 2 $\leq$ x or -1<x<1

ugcnetcse-july2016-paper3 linear-programming

Answer key

13.8 Number Representation (1)

13.8.1 Number Representation: UGC NET CSE | December 2011 | Part 2 | Question: 34

Negative numbers cannot be represented in

A. Signed magnitude form
B. $1's$ complement form
C. $2's$ complement form
D. None of the above

ugcnetcse-dec2011-paper2 discrete-mathematics number-representation

Answer key

13.9 Partial Order (2)

13.9.1 Partial Order: UGC NET CSE | December 2010 | Part 2 | Question: 3

A partially ordered set is said to be a lattice if every two elements in the set have

A. A unique least upper bound
B. A unique greatest lower bound
C. Both (A) and (B)
D. None of the above

ugcnetcse-dec2010-paper2 discrete-mathematics partial-order

Answer key

13.9.2 Partial Order: UGC NET CSE | July 2018 | Part 2 | Question: 90

Which of the following statements is true?

A. $(Z, \leq)$ is not totally ordered
B. The set inclusion relation $\subseteq$ is a partial ordering on the power set of a set S
C. $(Z, \neq)$ is a poset
D. The directed graph  is not a partial order

ugcnetcse-july2018-paper2 discrete-mathematics partial-order

Answer key

13.10 Propositional Logic (8)

13.10.1 Propositional Logic: UGC NET CSE | August 2016 | Part 2 | Question: 2

Let us assume that you construct ordered tree to represent the compound proposition

$(\sim (p \wedge q)) \leftrightarrow (\sim p \vee \sim q)$.

Then, the prefix expression and post-fix expression determined using this ordered tree are given as _____ and ______ respectively.

A. $\leftrightarrow \sim \wedge pq \vee \sim\sim pq, pq \wedge \sim p \sim q \sim \vee \leftrightarrow$
B. $\leftrightarrow \sim \wedge pq \vee \sim p \sim q, pq \wedge \sim p \sim q \sim \vee \leftrightarrow$
C. $\leftrightarrow \sim \wedge pq \vee \sim\sim pq, pq \wedge \sim p \sim\sim q \vee \leftrightarrow$
D. $\leftrightarrow \sim \wedge pq \vee \sim p \sim q, pq \wedge \sim p \sim\sim q \vee \leftrightarrow$

ugcnetcse-aug2016-paper2 discrete-mathematics propositional-logic

Answer key

13.10.2 Propositional Logic: UGC NET CSE | August 2016 | Part 3 | Question: 70

Let $\nu(x)$ mean x is a vegetarian, $m(y)$ for y is meat, and $e(x,y)$ for x eats y. Based on these, consider the following sentences :

I. $\forall x \vee (x) \Leftrightarrow (\forall y e(x,y) \implies \neg m(y))$

II.$\forall x \vee (x) \Leftrightarrow (\neg(\exists y m(y) \wedge e(x,y)))$

III.$\forall x(\exists y m(y) \wedge e(x,y)) \Leftrightarrow (x) \Leftrightarrow \neg \vee (x)$

One can determine that

A. Only I and II are equivalent sentences
B. Only II and III are equivalent sentences.
C. Only I and III are equivalent sentence .
D. I, II, and III are equivalent sentences.

ugcnetcse-aug2016-paper3 discrete-mathematics propositional-logic

Answer key

13.10.3 Propositional Logic: UGC NET CSE | August 2016 | Part 3 | Question: 74

Consider the following logical inferences :

I_1 : If it is Sunday then school will not open.

The school was open.

Inference : It was not Sunday.

I_2 : If it is Sunday then school will not open.

It was not Sunday.

Inference : The school was open.

Which of the following is correct ?

A. Both I_1 and I_2 are correct inferences.
B. I_1 is correct but I_2 is not a correct inference.
C. I_1 is not correct but I_2 is a correct inference.
D. Both I_1 and I_2 are not correct inferences.

ugcnetcse-aug2016-paper3 discrete-mathematics propositional-logic

Answer key

13.10.4 Propositional Logic: UGC NET CSE | December 2011 | Part 2 | Question: 43

The proposition $\sim$ qvp is equivalent to

A. p $\rightarrow$ q
B. q $\rightarrow$ p
C. p $\leftrightarrow$ q
D. p $\vee$ q

ugcnetcse-dec2011-paper2 discrete-mathematics propositional-logic

Answer key

13.10.5 Propositional Logic: UGC NET CSE | December 2012 | Part 3 | Question: 75

Let $\theta(x, y, z)$ be the statement "x+y=z" and let there be two quantification given as

I. $\forall x \forall y \exists z \theta(x, y, z)$
II. $\exists z \forall x \forall y \theta(x, y, z)$

where x, y, z are real numbers, then which one of the following is correct?

A. I is true and II is true
B. I is true and II is false
C. I is false and II is true
D. I is false and II is false

ugcnetcse-dec2012-paper3 discrete-mathematics propositional-logic

Answer key

13.10.6 Propositional Logic: UGC NET CSE | December 2015 | Part 2 | Question: 8

Match the following :

	List-I		List-II
(a)	Vacuous proof	(i)	A proof that the implication $p \to q$ is true based on the fact that p is false.
(b)	Trivial proof	(ii)	A proof that the implication $p \to q$ is true based on the fact that q is true.
(c)	Direct proof	(iii)	A proof that the implication $p \to q$ is true that proceeds by showing that q must be true when p is true.
(d)	Indirect proof	(iv)	A proof that the implication $p \to q$ is true that proceeds by showing that p must be false when q is false.

Codes :

A. (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
B. (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
C. (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
D. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

ugcnetcse-dec2015-paper2 discrete-mathematics propositional-logic

Answer key

13.10.7 Propositional Logic: UGC NET CSE | July 2018 | Part 2 | Question: 76

Consider the following statements:

a. False $\models$ True
b. If $\alpha \models (\beta \wedge \gamma$ then $\alpha \models \gamma$

Which of the following is correct with respect to above statements?

A. Both statement a and statement b are false
B. Statement a is true and statement b is false
C. Statement a is false and statement b is true
D. Both statement a and statement b are true

ugcnetcse-july2018-paper2 discrete-mathematics propositional-logic

Answer key

13.10.8 Propositional Logic: UGC NET CSE | Junet 2015 | Part 3 | Question: 57

In propositional language $P \leftrightarrow Q$ is equivalent to (where $\sim$ denotes NOT)

A. $\sim (P \vee Q) \wedge \sim (Q \vee P)$
B. $(\sim P \vee Q) \wedge (\sim Q \vee P)$
C. $(P \vee Q) \wedge (Q \vee P)$
D. $\sim (P \vee Q) \to \sim (Q \vee P)$

ugcnetcse-june2015-paper3 discrete-mathematics propositional-logic

Answer key

13.11 Quantifiers (1)

13.11.1 Quantifiers: UGC NET CSE | July 2018 | Part 2 | Question: 85

The equivalence of $\neg \exists x\, Q(x)$ is

A. $\exists x \neg Q(x)$
B. $\forall x \neg Q(x)$
C. $\neg \exists x \neg Q(x)$
D. $\forall x\, Q(x)$

ugcnetcse-july2018-paper2 discrete-mathematics quantifiers

Answer key

13.12 Relations (1)

13.12.1 Relations: UGC NET CSE | July 2018 | Part 2 | Question: 88

Which of the relations on $\{0, 1, 2, 3\}$ is an equivalence relation?

A. { (0, 0) (0, 2) (2, 0) (2, 2) (2, 3) (3, 2) (3, 3) }
B. { (0, 0) (1, 2) (2, 2) (3, 3) }
C. { (0, 0) (0, 1) (0, 2) (1, 0) (1, 1) (1, 2) (2, 0) }
D. { (0, 0) (0, 2) (2, 3) (1, 1) (2, 2)

ugcnetcse-july2018-paper2 discrete-mathematics relations

Answer key

13.13 Set Theory (2)

13.13.1 Set Theory: UGC NET CSE | August 2016 | Part 2 | Question: 3

Let A and B be sets in a finite universal set U. Given the following : $|A-B|, |A \bigoplus B|, |A|+|B|$ and $|A \cup B|$ Which of the following is in order of increasing size ?

A. $|A-B| < |A \bigoplus B| < |A|+|B| < |A \cup B|$
B. $|A \bigoplus B| < |A-B| < |A \cup B| < |A|+|B|$
C. $|A \bigoplus B| < |A|+|B| < |A-B| < |A \cup B|$
D. $|A-B| < |A \bigoplus B| < |A \cup B| < |A|+|B|$

ugcnetcse-aug2016-paper2 discrete-mathematics set-theory

Answer key

13.13.2 Set Theory: UGC NET CSE | December 2012 | Part 3 | Question: 34

The power set of AUB, where A={2, 3, 5, 7} and B={2, 5, 8, 9}is

A. 256 B. 64 C. 16 D. 4

ugcnetcse-dec2012-paper3 engineering-mathematics discrete-mathematics set-theory

Answer key

Answer Keys

13.0.1	B	13.0.2	D	13.0.3	C	13.0.4	B	13.0.5	B
13.0.6	Q-Q	13.0.7	C	13.0.8	A	13.0.9	B	13.0.10	D
13.0.11	D	13.1.1	Q-Q	13.2.1	D	13.2.2	B	13.3.1	Q-Q

13.4.1	C
13.7.1	A
13.10.2	Q-Q
13.10.7	Q-Q
13.13.2	B

13.4.2	B
13.8.1	Q-Q
13.10.3	Q-Q
13.10.8	B

13.5.1	Q-Q
13.9.1	Q-Q
13.10.4	Q-Q
13.11.1	Q-Q

13.5.2	C
13.9.2	Q-Q
13.10.5	B
13.12.1	Q-Q

13.6.1	C
13.10.1	Q-Q
13.10.6	A
13.13.1	D